Derivatives: Contracts, Models, and Markets · Technical Note 21

Local Volatility, Heston's Characteristic Function, and the Leverage Function

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Chapter 21 of Derivatives: Contracts, Models, and Markets states three results and derives none of them: that Dupire's formula reads a unique local volatility off an arbitrage-free call surface, that Heston's characteristic function is available in closed form, and that the leverage function of a local-stochastic model is defined by a fixed point. That is the right division for a chapter whose subject is what each model captures and where it fails. This note supplies the three derivations, together with the two lines that turn Heston's characteristic function into Bates's, and shows that the three are one derivation seen from three sides: Gyöngy's theorem is what makes Dupire's function a conditional average, which is why a local-volatility surface is twice as steep as the implied surface, and which is also why the leverage function's defining equation is circular. It then gives the scaling argument behind the short-maturity skew (the failure that motivated everything after Section 21.7), and shows that jumps change the mechanism of that failure without curing it. A last section, moved from Chapter 14, records why fractional Brownian motion, the driver of rough volatility, cannot itself price options: the fractional Black–Scholes formula, the failure of hedging on both sides of $H = 1/2$, and the drift of a simulated fractional path.

1 DUPIRE'S FORMULA

Take a single asset with $r = q = 0$; the general case is restored at the end and changes nothing structural. Let the asset follow

$$dS = \sigma_{\text{loc}}(S,t)\, S\, dW \tag{TN21.1}$$

and let $p(x,T)$ be the density of S_T. The density solves the **forward Kolmogorov equation**

$$\frac{\partial p}{\partial T} = \frac{1}{2}\frac{\partial^2}{\partial x^2}\Big[\sigma_{\text{loc}}^2(x,T)\, x^2\, p(x,T)\Big] \tag{TN21.2}$$

which is the statement that probability is conserved as it diffuses—the adjoint of the backward equation that prices options.

Write the call price as an integral against that density,

$$C(K,T) = \int_K^\infty (x-K)\, p(x,T)\, dx \tag{TN21.3}$$

Two derivatives of (TN21.3) *in K give the density, and one in T gives the numerator.* Differentiating in K under the integral,

$$\frac{\partial C}{\partial K} = -\int_K^\infty p(x,T)\, dx, \qquad \frac{\partial^2 C}{\partial K^2} = p(K,T) \tag{TN21.4}$$

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which is Breeden–Litzenberger, and is the reason Section 21.2 can call the denominator of Dupire's formula "the implied density."

Now differentiate (TN21.3) in T and substitute (TN21.2):

$$\frac{\partial C}{\partial T} = \int_K^\infty (x-K)\,\frac{\partial p}{\partial T}\,dx = \frac{1}{2}\int_K^\infty (x-K)\,\frac{\partial^2}{\partial x^2}\Big[\sigma_{\text{loc}}^2 x^2 p\Big]\,dx \tag{TN21.5}$$

Integrate by parts twice, writing $\Psi(x) = \sigma_{\text{loc}}^2(x,T)x^2p(x,T)$ for brevity. The first integration gives $\big[(x-K)\Psi'(x)\big]_K^\infty - \int_K^\infty \Psi'(x)\,dx$; the boundary term vanishes at $x = K$ because $(x-K) = 0$ there, and at infinity because a density with a finite mean decays faster than Ψ' grows. The second gives $-\big[\Psi(x)\big]_K^\infty = \Psi(K)$. Hence

$$\frac{\partial C}{\partial T} = \frac{1}{2}\,\sigma_{\text{loc}}^2(K,T)\,K^2\,p(K,T) = \frac{1}{2}\,\sigma_{\text{loc}}^2(K,T)\,K^2\,\frac{\partial^2 C}{\partial K^2} \tag{TN21.6}$$

and solving for σ_{loc} gives the chapter's equation (21.2) with $r = q = 0$.

Restoring the drift. Two things change and they interact. The forward equation gains a transport term,

$$\frac{\partial p}{\partial T} = -\frac{\partial}{\partial x}\Big[(r-q)\,x\,p(x,T)\Big] + \frac{1}{2}\frac{\partial^2}{\partial x^2}\Big[\sigma_{\text{loc}}^2(x,T)\,x^2\,p(x,T)\Big] \tag{TN21.7}$$

and the call is now a discounted expectation, $C(K,T) = e^{-rT}I(K,T)$ with $I = \int_K^\infty (x-K)\,p\,dx$. The discount factor carries into both derivatives: $\partial C/\partial K = -e^{-rT}\int_K^\infty p\,dx$ and $\partial^2 C/\partial K^2 = e^{-rT}p(K,T)$ replace (TN21.4), and in T

$$\frac{\partial C}{\partial T} = -rC + e^{-rT}\frac{\partial I}{\partial T} \tag{TN21.8}$$

The diffusion term of (TN21.7) integrates exactly as before, to $\frac{1}{2}\sigma_{\text{loc}}^2(K,T)K^2p(K,T)$. The transport term needs only *one* integration by parts, the boundary term vanishing at $x = K$ because $(x-K) = 0$ and at infinity because the mean is finite:

$$-\int_K^\infty (x-K)\,\frac{\partial}{\partial x}\Big[(r-q)\,x\,p\Big]dx = (r-q)\int_K^\infty x\,p\,dx = (r-q)\left(I - Ke^{rT}\frac{\partial C}{\partial K}\right)$$

the last step splitting $x = (x-K)+K$ and using the K-derivative above. Substituting both terms into (TN21.8),

$$\frac{\partial C}{\partial T} = -rC + (r-q)C - (r-q)K\frac{\partial C}{\partial K} + \frac{1}{2}\sigma_{\text{loc}}^2(K,T)\,K^2\frac{\partial^2 C}{\partial K^2} \tag{TN21.9}$$

The two rate terms collapse: $-rC + (r-q)C = -qC$, which is where the qC of the numerator comes from—it is the discounting and the drift not quite canceling, not a dividend term carried along from the start. Moving everything but the second derivative to the left,

$$\frac{1}{2}\sigma_{\text{loc}}^2(K,T)\,K^2\frac{\partial^2 C}{\partial K^2} = \frac{\partial C}{\partial T} + (r-q)K\frac{\partial C}{\partial K} + qC$$

which is the chapter's equation (21.2). ■

Three remarks, each of which is a place where the formula is over-read.

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Nothing has been assumed about how the surface was generated. The derivation uses only that C came from *some* density and that the density evolves by (TN21.2). If the surface came from a stochastic-volatility model, from a jump model, or from a trader's marks, Dupire's formula still returns the unique one-factor diffusion reproducing it.

Existence is exactly no-arbitrage. The formula requires $\partial^2 C/\partial K^2 > 0$ and $\partial C/\partial T > 0$. The first is Section 19.5's butterfly condition, the second its calendar condition. A surface violating either does not merely calibrate badly; it has no local-volatility model at all, because the "density" it implies is not one.

The formula is a second derivative of a price, and that is a numerical problem, not a conceptual one. Interpolating quoted prices and differencing them twice amplifies quote noise brutally. Practical implementations difference a *smooth parameterization* of the implied-volatility surface instead, or work in log-strike, where $K^2\partial^2 C/\partial K^2 = C_{yy} - C_y$ with $y = \log K$ and the wings behave.

2 GYÖNGY'S THEOREM, AND WHY LOCAL VOLATILITY IS AN AVERAGE

Section 1 says what σ_{loc} *is* as a function of the surface. It does not say what it *means* when the true dynamics are not a one-factor diffusion. The following does, and it is the hinge of the whole chapter.

> **Gyöngy's theorem.** Let S satisfy $dS = \sigma_t S\, dW$, where σ_t is any adapted process with $\mathbb{E}\int_0^T \sigma_t^2\, dt < \infty$—stochastic, path-dependent, driven by any number of factors. Define
>
> $$\hat{\sigma}^2(x,t) = \mathbb{E}\big[\,\sigma_t^2 \;\big|\; S_t = x\,\big] \qquad \text{(TN21.10)}$$
>
> Then the one-factor diffusion $d\hat{S} = \hat{\sigma}(\hat{S},t)\hat{S}\, dW$, started at $\hat{S}_0 = S_0$, has the same *marginal* distribution as S at every fixed t.

The reason is (TN21.2) again. The marginal density of S satisfies a forward equation whose diffusion coefficient is $\mathbb{E}[\sigma_t^2 \mid S_t = x]x^2$—the conditioning arises because the equation governs the marginal, and everything the extra factors do enters only through their average given the current price. The one-factor model with that coefficient therefore solves the same equation with the same initial condition. ■

Two consequences, and between them they carry Sections 21.2, 21.3 and 21.9 of the chapter.

Dupire's function is a conditional average. Since a call price depends on the marginal alone, matching the surface means matching marginals, so

$$\sigma_{\text{loc}}^2(x,t) = \mathbb{E}\big[\,\sigma_t^2 \;\big|\; S_t = x\,\big] \qquad \text{(TN21.11)}$$

This is the averaging argument of Example 21.1 stated exactly. An implied volatility averages over paths; the local volatility at (x,t) averages over everything the true model was doing at the moment it passed through x. Both are averages, taken over different things, and the local one is the less averaged of the two—which is why it is the steeper.

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It also explains why Example 21.1's ratio is 1.75 and not exactly 2. The factor of two is the leading term of a small-maturity, small-skew expansion in which the implied variance at strike K is approximately the average of the local variance over the interval between S_0 and K; a linear local-variance function then gives an implied variance sloping at exactly half the rate. Everything that makes the surface not locally linear—curvature, maturity, the wings—moves the realized ratio off two, and the direction is not fixed.

Matching the surface constrains only marginals. This is the sharpest thing in the note, and the whole of Section 21.3 follows from it. Two models with identical marginals at every t agree on every European option and may disagree on everything else, because a marginal says nothing about the *joint* law across dates. A forward-starting option, a cliquet, a barrier and an American option all depend on the joint law. *So the exactness of the local-volatility fit and the wrongness of its forward smile are not in tension; they are the same fact.*

3 HESTON'S CHARACTERISTIC FUNCTION

Write $x_t = \log S_t$ and work under the risk-neutral measure. *Set $r = q = 0$ for the derivation*; the drift they contribute is deterministic, and the end of this section puts it back exactly. Chapter 21's equation (21.4) becomes

$$dx = -\frac{1}{2}v\,dt + \sqrt{v}\,dW_1, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2, \qquad dW_1 dW_2 = \rho\,dt \tag{TN21.12}$$

We want $\varphi(u) = \mathbb{E}\big[e^{iux_T}\big]$. Define $f(x, v, t) = \mathbb{E}\big[e^{iux_T} \mid x_t = x,\ v_t = v\big]$, which by Feynman–Kac satisfies

$$\frac{\partial f}{\partial t} + \frac{1}{2}v\frac{\partial^2 f}{\partial x^2} - \frac{1}{2}v\frac{\partial f}{\partial x} + \rho\sigma_v v\frac{\partial^2 f}{\partial x\,\partial v} + \frac{1}{2}\sigma_v^2 v\frac{\partial^2 f}{\partial v^2} + \kappa(\theta - v)\frac{\partial f}{\partial v} = 0 \tag{TN21.13}$$

with terminal condition $f(x, v, T) = e^{iux}$.

The model is affine, and that is the entire reason it is tractable. Every coefficient in (TN21.13) is affine in the state (x, v), which makes the following ansatz consistent rather than lucky. Write $\tau = T - t$ and try

$$f = \exp\big\{\, iux + A(\tau) + B(\tau)\,v \,\big\}, \qquad A(0) = B(0) = 0 \tag{TN21.14}$$

Substituting and dividing through by f—each derivative brings down a factor, so $f_x = iuf$, $f_{xx} = -u^2 f$, $f_v = Bf$, $f_{vv} = B^2 f$, $f_{xv} = iuBf$, and $f_t = -(A' + B'v)f$—gives

$$-(A' + B'v) - \frac{1}{2}vu^2 - \frac{1}{2}iuv + \rho\sigma_v v\,iuB + \frac{1}{2}\sigma_v^2 vB^2 + \kappa(\theta - v)B = 0 \tag{TN21.15}$$

This must hold for every v, so the terms proportional to v and the terms free of v vanish separately. Writing $b = \kappa - \rho\sigma_v iu$, that is

$$B' = \frac{1}{2}\sigma_v^2 B^2 - bB - \frac{1}{2}\big(u^2 + iu\big), \qquad A' = \kappa\theta B \tag{TN21.16}$$

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The first is a Riccati equation with constant coefficients, which is why a closed form exists at all; the second is then an integral. Setting

$$d = \sqrt{b^2 + \sigma_v^2\big(u^2 + iu\big)}\,, \qquad g = \frac{b-d}{b+d} \tag{TN21.17}$$

the solutions satisfying $A(0) = B(0) = 0$ are

$$B(\tau) = \frac{b-d}{\sigma_v^2} \cdot \frac{1 - e^{-d\tau}}{1 - g e^{-d\tau}}, \qquad A(\tau) = \frac{\kappa\theta}{\sigma_v^2}\left[(b-d)\tau - 2\log\frac{1 - g e^{-d\tau}}{1-g}\right] \tag{TN21.18}$$

and the characteristic function of the log price is

$$\varphi(u) = \exp\Big\{ iu\big(\log S_0 + (r-q)T\big) + A(T) + B(T)v_0 \Big\} \tag{TN21.19}$$

■

The $(r-q)T$ has reappeared in a derivation that set $r = q = 0$, and the restoration is exact rather than approximate. Neither rate enters the variance dynamics, so normalizing them away changes nothing about v; and the drift they contribute to x is deterministic. Driving the general and the normalized model with the same Brownian paths therefore gives, path by path,

$$x_T = x_T^{(0)} + (r-q)T$$

with $x^{(0)}$ the driftless log price of (TN21.12). Taking the expectation of e^{iux_T} multiplies the characteristic function by the constant $e^{iu(r-q)T}$, which is the term (TN21.19) carries. *It is also why the Riccati system never sees r or q*—a deterministic shift cannot reach it.

The branch, and the trap. Both d and the logarithm in (TN21.18) are complex, and d is defined only up to sign. Choosing $-d$ instead of d gives an algebraically equivalent expression—it replaces g by $1/g$ and $e^{-d\tau}$ by $e^{+d\tau}$—which is *not* numerically equivalent. *With the sign as written, $|g| < 1$ and $|e^{-d\tau}| \le 1$, so $1 - ge^{-d\tau}$ stays in the right half plane and the principal logarithm is continuous in τ and in u.* With the other sign the argument of the logarithm crosses the negative real axis as τ or u grows, the principal branch jumps by $2\pi i$, and the computed characteristic function acquires discontinuities that destroy the pricing integral. This is the *little Heston trap*: models that "fail to calibrate at long maturities" are, often enough, this and nothing else.

4 ADDING JUMPS: THE BATES CHARACTERISTIC FUNCTION

Chapter 21's equation (21.7) appends Merton's jumps to Heston's price equation and states that the characteristic function becomes a product. In log terms, again with $r = q = 0$,

$$dx = -\Big(\frac{1}{2}v + \lambda\bar{k}\Big)dt + \sqrt{v}\,dW_1 + \log Y\,dN, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2 \tag{TN21.20}$$

where N is a Poisson process of intensity λ, the jump sizes $\log Y$ are independent draws from $N(\mu_J, \delta^2)$, and N and the Y are independent of both Brownian motions.

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The chapter parameterizes the jump by its mean effect on the price, $\bar{k} = \mathbb{E}[Y] - 1 = e^{\mu_J + \delta^2/2} - 1$, so

$$\mu_J = \log(1 + \bar{k}) - \frac{\delta^2}{2} \tag{TN21.21}$$

which is the relation quoted beside the chapter's formula.

The compensator. The $-\lambda\bar{k}$ in the drift is what keeps $S_t = e^{x_t}$ a martingale. Over $[0, t]$ the jumps multiply the price by $\prod_{i \le N_t} Y_i$, and conditioning on the number of jumps,

$$\mathbb{E}\Big[\prod_{i \le N_t} Y_i\Big] = \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} (1 + \bar{k})^n = e^{\lambda \bar{k} t}$$

The continuous part contributes $\mathbb{E}\big[\exp\big(\int_0^t \sqrt{v}\, dW_1 - \tfrac{1}{2}\int_0^t v\, ds\big)\big] = 1$, the exponential martingale, and independence lets the two expectations multiply: $\mathbb{E}[S_t] = S_0\, e^{\lambda\bar{k}t}\, e^{-\lambda\bar{k}t} = S_0$. Without the compensator a model with negative mean jumps would carry the price below the forward, which is the chapter's remark.

Independence turns the sum into a product. Write the terminal log price as

$$x_T = x_T^{(H)} + J_T - \lambda\bar{k}T, \qquad J_T = \sum_{i=1}^{N_T} \log Y_i$$

where $x^{(H)}$ is Heston's log price of (TN21.12) and J_T is the compound Poisson sum. J_T is independent of $x_T^{(H)}$, so the expectation of e^{iux_T} factorizes,

$$\varphi_{\text{Bates}}(u) = \mathbb{E}\big[e^{iux_T^{(H)}}\big]\, \mathbb{E}\big[e^{iuJ_T}\big]\, e^{-iu\lambda\bar{k}T}$$

and the middle factor is elementary. Conditional on $N_T = n$ the sum of n independent normals has characteristic function $\varphi_Y(u)^n$, with $\varphi_Y(u) = \mathbb{E}\big[e^{iu\log Y}\big] = \exp\big(iu\mu_J - u^2\delta^2/2\big)$; averaging over the Poisson weights,

$$\mathbb{E}\big[e^{iuJ_T}\big] = \sum_{n=0}^{\infty} e^{-\lambda T} \frac{(\lambda T)^n}{n!} \varphi_Y(u)^n = \exp\big\{\lambda T\big(\varphi_Y(u) - 1\big)\big\} \tag{TN21.22}$$

Hence

$$\varphi_{\text{Bates}}(u) = \varphi_{\text{Heston}}(u) \cdot \exp\Big\{\lambda T\big(e^{\,iu\mu_J - u^2\delta^2/2} - 1\big) - iu\lambda\bar{k}T\Big\} \tag{TN21.23}$$

which is the chapter's formula. ■ The inversion of Section 5 then prices the model unchanged, including the normalization $\varphi(-i) = S_0 e^{(r-q)T}$: at $u = -i$ the jump factor is $\exp\{\lambda T(\varphi_Y(-i) - 1) - \lambda\bar{k}T\}$ and $\varphi_Y(-i) = e^{\mu_J + \delta^2/2} = 1 + \bar{k}$, so the factor is exactly one—which is the compensator doing its job a second time.

The same result from the pricing equation, and what it shows. The Feynman–Kac equation (TN21.13) gains two terms: the drift $-\lambda\bar{k}\, \partial f/\partial x$, and the jump operator

$$\lambda \int \big[f(x + y, v, t) - f(x, v, t)\big]\, \nu(dy)$$

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where ν is the $N(\mu_J, \delta^2)$ law of a jump. Under the affine ansatz (TN21.14), $f(x + y, v, t) = e^{iuy} f(x, v, t)$, so the integral is $\lambda\big(\varphi_Y(u) - 1\big) f$—a constant multiple of f with no v in it. Collecting terms as in (TN21.15), the terms proportional to v are untouched and the terms free of v gain $\lambda(\varphi_Y(u) - 1) - iu\lambda\bar{k}$:

$$B' = \frac{1}{2}\sigma_v^2 B^2 - bB - \frac{1}{2}\big(u^2 + iu\big), \qquad A' = \kappa\theta B + \lambda\big(\varphi_Y(u) - 1\big) - iu\lambda\bar{k} \quad \text{(TN21.24)}$$

B is Heston's, and A gains a term linear in τ: $A_{\text{Bates}}(\tau) = A_{\text{Heston}}(\tau) + \tau\big[\lambda(\varphi_Y(u) - 1) - iu\lambda\bar{k}\big]$. Exponentiating recovers (TN21.23).

The jumps enter A and not B because neither their intensity nor their size depends on v. A state-independent jump is an affine term whose coefficient on the state is zero, and the Riccati equation for B never sees it. That is also the boundary of the trick. Let the intensity be $\lambda_0 + \lambda_1 v$—jumps that arrive more often when volatility is high, which is what the data show—and the model is still affine, but $\lambda_1(\varphi_Y(u) - 1)v$ now sits among the v-terms, B's equation changes, and the characteristic function is no longer Heston's times anything. The same happens if the variance itself jumps. Duffie, Pan, and Singleton (2000) give the general affine machinery, of which Bates is the case that factorizes; the chapter's remark that the independence "is what makes the model usable" is the statement that in this case nothing beyond Heston's Riccati solution has to be solved.

Merton's series is the same formula with the variance frozen. Set $\sigma_v = 0$ and $v_0 = \theta = \sigma^2$, so that the variance is constant. Then $B' = -\kappa B - \frac{1}{2}(u^2 + iu)$ gives $B(\tau) = -\frac{1}{2}(u^2 + iu)(1 - e^{-\kappa\tau})/\kappa$ and $A = \kappa\sigma^2 \int_0^\tau B$, and the two combine into $A(\tau) + B(\tau)\sigma^2 = -\frac{1}{2}(u^2 + iu)\sigma^2\tau$—the mean reversion cancels out of a variance that never leaves θ. Equation (TN21.23) becomes

$$\varphi_{\text{Merton}}(u) = \exp\left\{iux_0 - \frac{1}{2}\sigma^2 T\big(u^2 + iu\big) - iu\lambda\bar{k}T\right\} \cdot e^{-\lambda T} \sum_{n=0}^{\infty} \frac{(\lambda T)^n}{n!}\, \varphi_Y(u)^n$$

where the jump factor has been written back as the series (TN21.22) rather than the exponential. The n-th term has exponent $iu\big(x_0 - \lambda\bar{k}T + n\mu_J\big) - \frac{1}{2}u^2\big(\sigma^2 T + n\delta^2\big) - \frac{1}{2}iu\sigma^2 T$, and substituting $n\mu_J = n\log(1 + \bar{k}) - n\delta^2/2$ from (TN21.21) rearranges it to

$$iu\Big(x_0 + r_n T - \frac{1}{2}\sigma_n^2 T\Big) - \frac{1}{2}u^2\sigma_n^2 T, \qquad \sigma_n^2 = \sigma^2 + \frac{n\delta^2}{T}, \quad r_n = -\lambda\bar{k} + \frac{n\log(1 + \bar{k})}{T}$$

which is the characteristic function of a lognormal price with volatility σ_n and drift r_n. Conditional on n jumps, then, the option is a Black–Scholes call with those inputs, and the only care needed is the discounting: the Black–Scholes–Merton formula discounts at r_n where the true rate is zero, so each term carries a correction $e^{r_n T} = e^{-\lambda\bar{k}T}(1 + \bar{k})^n$ that folds into the Poisson weight,

$$e^{-\lambda T}\frac{(\lambda T)^n}{n!}\, e^{-\lambda\bar{k}T}(1 + \bar{k})^n = e^{-\lambda^* T}\frac{(\lambda^* T)^n}{n!}, \qquad \lambda^* = \lambda(1 + \bar{k})$$

This is Section 21.5's series, and it explains the remark there that the intensity is λ^* rather than λ "because the sum discounts at r while each term discounts at its own r_n."

■

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5 FROM THE CHARACTERISTIC FUNCTION TO A PRICE

The characteristic function determines the distribution, and the map back is Gil-Pelaez inversion. For the digital probability,

$$P_2 = \mathbb{Q}\big(S_T > K\big) = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\log K}\varphi(u)}{iu}\right] du \tag{TN21.25}$$

The delta term is the same probability under the measure that uses the asset as numeraire, which re-weights the characteristic function by S_T and therefore evaluates it one unit down the imaginary axis:

$$P_1 = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\log K}\varphi(u-i)}{iu\,\varphi(-i)}\right] du, \qquad \varphi(-i) = S_0 e^{(r-q)T} \tag{TN21.26}$$

and the call is $C = S_0 e^{-qT} P_1 - K e^{-rT} P_2$.

The integrand is smooth and decays exponentially, because $|\varphi(u)|$ falls off like e^{-cu} for a diffusion. A fixed Gauss–Legendre rule on a truncated interval is therefore not a compromise: the accompanying notebook uses 512 nodes on $[0, 150]$ and agrees with adaptive quadrature to about 3×10^{-9}, uniformly in strike. That precision is not vanity. It is what allows the *second* difference of (TN21.6) to be taken numerically at all, which is how the notebook derives a local-volatility surface from Heston's own prices—the construction Example 21.7 rests on.

Two practical points. The integrand at $u \to 0$ is finite but is written as a ratio with iu in the denominator, so a quadrature rule with a node at zero must be avoided rather than patched; Gauss–Legendre never places one. And prices for many strikes share the same $\varphi(u)$ values, so a whole smile costs barely more than one option—which is what makes calibrating to a surface in seconds possible, and is the real reason Heston is a workhorse.

6 THE LEVERAGE FUNCTION AND ITS FIXED POINT

The local-stochastic model of Chapter 21's equation (21.10) is

$$dS = (r-q)S\,dt + L(S,t)\sqrt{v}\,S\,dW_1 \tag{TN21.27}$$

with v a stochastic-variance process. The instantaneous volatility is $\sigma_t = L(S_t,t)\sqrt{v_t}$, so (TN21.11) says the model reproduces the vanilla surface if and only if

$$\mathbb{E}\Big[L^2(S_t,t)\,v_t \;\Big|\; S_t = x\Big] = \sigma_{\mathrm{loc}}^2(x,t) \tag{TN21.28}$$

Because $L(S_t,t)$ is a deterministic function of the conditioning variable, it comes out of the expectation, and

$$\boxed{L^2(x,t) = \frac{\sigma_{\mathrm{loc}}^2(x,t)}{\mathbb{E}\big[\,v_t \mid S_t = x\,\big]}} \tag{TN21.29}$$

This is the fixed point. The right-hand side is not data: the conditional expectation is taken under the law of (S, v), and that law depends on L through (TN21.27). The equation defines L in terms of a quantity that cannot be computed until L is known.

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Three things follow, and the first two are the reasons the construction is worth its cost.

The two extremes are limits of one model. Scaling the vol-of-vol by a mixing parameter $\zeta \in [0,1]$ interpolates. At $\zeta = 0$ the variance process is deterministic, $\mathbb{E}[v_t \mid S_t = x] = v_t$ is independent of x, and (TN21.29) makes L proportional to σ_{loc}: pure local volatility. At $\zeta = 1$ the stochastic part explains as much of the smile as it can and L absorbs only the residual. *Every setting reproduces the vanilla surface exactly*, which is what makes the parameter a measure of model risk rather than a fit quality knob—the notebook confirms it at $\zeta = 0$, 0.5 and 1, and prices the 110 up-and-out of Example 21.7 at 0.420, 0.572 and 0.845 across that range.

Two ways to break the circle. The **forward-PDE** method discretizes the joint density $p(x, v, t)$ and steps it forward; at each time step $\mathbb{E}[v \mid S = x]$ is read off the density just computed, L is updated on the slice, and the step is taken. The **particle** method simulates a cloud of paths and estimates the same conditional expectation from the cloud itself as it advances—by kernel regression, or by binning, which is what the notebook does. Both are the same idea: the fixed point is solved *one time step at a time*, because $L(\cdot, t)$ depends only on the law at t, which the simulation has already produced.

The estimator is where the accuracy goes. A conditional expectation estimated from finitely many paths is noisy exactly where paths are few, which is the wings—and the wings are where L is largest and the division in (TN21.29) least stable. Practical implementations bin on quantiles rather than on a fixed grid, floor the denominator, and smooth L across strikes before using it. A leverage function that looks jagged is not a discovery about the market.

7 WHY THE SHORT-MATURITY SKEW BROKE ALL THREE MODELS

Chapter 21 asserts two scalings without argument: that a classical stochastic-volatility model has $\psi(T) \to$ constant as $T \to 0$, and that a fractional driver with Hurst parameter H gives $\psi(T) \sim T^{H-1/2}$. Both follow from one observation. A third claim—that jumps repair the short end and lose the long end—follows from a different one, and the section closes with it.

To leading order in small T, the at-the-money skew of a stochastic-volatility model is governed by the covariance between the terminal log price and the realized variance along the way,

$$\psi(T) \;\propto\; \frac{1}{T}\,\mathrm{Cov}\left[\, x_T,\; \int_0^T v_s\, ds \,\right] \Big/ \text{(a variance scale)} \tag{TN21.30}$$

so everything turns on how fast that covariance vanishes relative to T.

For a diffusion it vanishes at exactly the same rate as T. The price accumulates variance at rate $O(T)$, and the variance process moves by $O(\sqrt{T})$ over the same window while the price moves by $O(\sqrt{T})$; their covariance over $[0, T]$ is therefore $O(T)$, the ratio is $O(1)$, and $\psi(T)$ tends to a finite constant. No choice of κ, σ_v or ρ changes the exponent, because none of them changes the fact that a Brownian increment scales as $\sqrt{T}$. *The saturation is structural, which is why fitting Heston to a whole surface means*

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fitting it badly everywhere.

A fractional driver changes the exponent because it changes the scaling. If log-variance is driven by a fractional Brownian motion with Hurst parameter H, its increment over $[0, T]$ scales as T^H rather than $T^{1/2}$. The covariance in (TN21.30) then scales as $T^{1/2} \cdot T^H \cdot T^{1/2} = T^{1+H-1/2}$, and dividing by the T of the denominator leaves

$$\psi(T) \;\sim\; T^{H-1/2} \tag{TN21.31}$$

which is Chapter 21's equation (21.8). Setting $H = 1/2$ recovers the constant, and $H \approx 0.1$ gives the observed $T^{-0.4}$.

Jumps change the mechanism, not the exponent at the short end. With a jump component the skew no longer comes from the covariance in (TN21.30), which is zero in Merton's model because the variance does not move; it comes from the asymmetry of the jump law itself. To leading order in the skewness of the return distribution, the at-the-money slope of the smile is $\psi(T) \approx -\gamma_1(T)/(6\sqrt{T})$, where γ_1 is the skewness of x_T (Backus, Foresi, and Wu, 2004). For a compound Poisson component the third cumulant is $\lambda T\, m_3$ with $m_3 = \mathbb{E}[(\log Y)^3] = \mu_J^3 + 3\mu_J\delta^2$, and the variance is $\sigma_{\text{tot}}^2 T$ with $\sigma_{\text{tot}}^2 = \sigma^2 + \lambda(\delta^2 + \mu_J^2)$, so

$$\gamma_1(T) = \frac{\lambda m_3}{\sigma_{\text{tot}}^3 \sqrt{T}}, \qquad \psi(T) \approx -\frac{\lambda m_3}{6\,\sigma_{\text{tot}}^3} \cdot \frac{1}{T} \tag{TN21.32}$$

The skew from jumps decays like $1/T$, which is faster than the market's $T^{-0.4}$ and faster than Heston's: the third cumulant grows only linearly with T while the variance scale it is measured against grows as $T^{3/2}$. This is Table 21.1's "skew dies too fast with maturity," and the approximation is already usable at a year: for Example 21.3's parameters, $\lambda m_3 = -0.00362$ and $\sigma_{\text{tot}} = 0.209$, so the slope $-\psi$ from (TN21.32) is -0.066 at one year and -0.022 at three, against exact values of -0.052 and -0.019.

At the short end the expansion fails—$\gamma_1 \to \infty$ as $T \to 0$ is outside the regime it is valid in—and the exact behavior is the opposite of a divergence. Over a window short enough that a jump is unlikely, $\lambda T \ll 1$, the region around the money is governed by the Brownian component alone, and the at-the-money slope tends to a finite limit exactly as it does for Heston: any process with a diffusive part saturates. Measured on the chapter's Merton case, the local exponent of $\psi(T)$ is -0.14 between one and two weeks, -0.75 between six months and a year, and -0.95 between two and three years, on its way to the -1 of (TN21.32). *So the jump model overshoots the one-month/one-year ratio not because it steepens faster than the market at the short end but because it decays faster at the long end*, and Bates, which takes its short-maturity skew from the jumps, inherits that decay—which is why Section 21.6 finds its three-year ratio almost exactly Merton's. Neither ingredient supplies a power law; the fractional driver does, because it changes the scaling of the increment itself rather than adding a second mechanism with its own, different, decay.

The economy of this is the argument for rough volatility, and should be stated as such. One parameter, estimated independently from high-frequency realized volatility under the physical measure, reproduces a cross-sectional feature of option prices under the risk-neutral measure that two decades of patching had not. The notebook checks

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the exponent directly: simulating rough Bergomi and regressing $\log \psi$ on $\log T$ over maturities from one month to one year returns slopes of -0.380, -0.281 and -0.066 at $H = 0.10$, 0.25 and 0.49, against the predicted -0.40, -0.25 and -0.01. Heston's own fitted slope over the same short end is -0.138.

8 FRACTIONAL BROWNIAN MOTION AS A PRICE

Section 14.13 of the book introduces fractional Brownian motion B_t^H, whose increments over an interval of length T have variance T^{2H} (equation (14.21)), and explains why it cannot model a traded price: its increments are correlated, so a strategy that trades on the correlation earns a profit with a risk that shrinks toward zero. Three questions the chapter leaves aside have answers worth recording: what the "fractional Black–Scholes" formula that is sometimes quoted says, why hedging fails on both sides of $H = 1/2$, and how a drift is added to a simulated fractional path.

Option prices. If a price followed $dS = \mu S\,dt + \sigma S\,dB_t^H$, then $\ln S_T$ would still be normal, with variance $\sigma^2 T^{2H}$ instead of $\sigma^2 T$, and the fractional Black–Scholes formula of Hu and Øksendal (2003) is the book's Chapter 15 formula with that one substitution: $\sigma\sqrt{T}$ becomes σT^H wherever it appears. Two things are wrong with treating the result as a price. The first is the arbitrage: there is no risk-neutral measure to take the expectation under, and the derivation reaches the formula through a calculus (the Wick product) in which the "self-financing portfolio" is not one any trader could hold, as Björk and Hult (2005) showed. The second is what the formula says about implied volatility. Substitute σT^H for $\sigma\sqrt{T}$ and the Black–Scholes–Merton implied volatility becomes $\sigma T^{H-1/2}$: at $H = 0.1$ and $\sigma = 20\%$ per year, an at-the-money option shows an implied volatility of 54% at one month and 13% at three years, with no smile at any maturity. No surface looks like that. The power law $T^{H-1/2}$ is real, but it belongs in the skew, as (TN21.31) has it, not in the level, and Chapter 21 shows that putting the roughness into volatility rather than into the price puts it where the market has it.

Hedging. The hedge breaks down too, in opposite ways on the two sides of one-half. Both trace to the squared increments: over a grid of step T/n they sum to about $T^{2H} n^{1-2H}$, where a Wiener path's sum to T. For $H > 1/2$ they sum to zero, so the Itô term of the book's Section 14.7 vanishes and any claim $f(S_T)$ is replicated by holding $f'(S_t)$ shares at a cost of $f(S_0)$: a call is replicated by holding one share whenever $S > K$ and none otherwise, from an outlay of $\max(S_0 - K, 0)$. The hedge is perfect, and that is the arbitrage: it says a call is worth its intrinsic value at every maturity. For $H < 1/2$ they sum to infinity, so the residual $\frac{1}{2}\,(\partial^2 f/\partial S^2)\,(\Delta S)^2$ that the delta hedge of Section 14.8 leaves behind grows without bound as the rebalancing gets finer, and no self-financing strategy replicates the payoff at any price. Completeness does not even arise: the question presupposes a risk-neutral measure, and there is none. The $H \approx 0.1$ measured in volatility would put a price on the unhedgeable side, one more reason the roughness belongs in volatility (a quantity no one replicates with) rather than in the price.

Adding a drift. The Cholesky construction of Section 14.13 builds a path with no

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drift, and adding one needs care. Because the variance over t is t^{2H} and not the sum of the per-step variances, a convexity correction added step by step does not add up to the right total, and the total changes when the step does. Add the drift as a function of the horizon, μt, and if the process is exponentiated add the convexity term $-\frac{1}{2}\sigma^2 t^{2H}$ that makes the mean come out as $e^{\mu t}$ (t^{2H}, not t). What μ should be is a different matter. For a price there is no risk-neutral answer, for the reason above: setting $\mu = r$ is a convention, not a consequence of anything. For volatility the question moves and is answered. In the rough model of Section 21.8 the price keeps the ordinary risk-neutral drift $r - q$, with q the dividend yield, because only its volatility is rough, and the variance's drift is pinned by the forward variance curve the market quotes, the prices today of variance delivered over each future period: the convexity term $-\frac{1}{2}\eta^2 t^{2H}$ is chosen so the expected variance at every horizon equals today's forward price of it.

9 WHERE THIS CAME FROM

Dupire's formula is Dupire (1994), with Derman and Kani (1994) giving the tree construction independently and at the same moment. Gyöngy (1986) is the source of Section 2's theorem, which predates the option-pricing application by a decade and is the reason so many apparently separate facts about local volatility turn out to be one fact. Heston (1993) derives the characteristic function; Albrecher, Mayer, Schoutens, and Tistaert (2007) named and diagnosed the branch problem of Section 3, and Lewis (2000) gives the single-integral pricing formula that many implementations prefer to the two-integral form used here. Bates (1996) introduced the model of Section 4 for Deutsche mark options; Duffie, Pan, and Singleton (2000) supply the affine framework in which its factorization is one case, and Merton (1976) the series it collapses to. The skewness expansion behind (TN21.32) is Backus, Foresi, and Wu (2004). The leverage-function construction is Ren, Madan, and Qian (2007) by forward PDE and Guyon and Henry-Labordère (2012) by particles; the latter also explains why the conditional expectation, not the calibration, is where the difficulty lives. The short-maturity scaling of Section 7 is made rigorous in Alòs, León, and Vives (2007); the empirical finding that $H \approx 0.1$ is Gatheral, Jaisson, and Rosenbaum (2018), and rough Bergomi is Bayer, Friz, and Gatheral (2016). For fractional Brownian motion as a price process, Rogers (1997) proves the arbitrage, Hu and Øksendal (2003) give the fractional Black–Scholes formula, Björk and Hult (2005) show why its self-financing portfolios are not ones a trader could hold, and Cheridito (2003) and Guasoni (2006) show the arbitrage vanishing under a minimal delay between trades and under proportional transaction costs respectively.

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